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Inventor(s)	Thorne; John

Reactive Target for Firearms Training

Abstract

A reactive target for firearms training is provided. The reactive target comprises: a body; an impact sensor configured to detect an impact on a target region of the body; an electromagnetic coupling arranged to support the body of the reactive target; and a controller configured to deactivate the electromagnetic coupling in response to the impact sensor detecting the impact for releasing the body of the reactive target to fall.

Inventors:	Thorne; John (Brackley, GB)
Applicant:	Thorney Limited (Brackley, GB)
Family ID:	1000007930551
Assignee:	Thorney Limited (Brackley, GB)
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Background/Summary

FIELD

[0001] The present specification relates to a reactive target for firearms training, and a corresponding kit and method.

BACKGROUND

[0002] With firearms training, there is a constant desire to have the training operation accurately reflect the real-world operational environment. This is particularly important for military and/or police training where critical skills need to be developed under realistic conditions.

[0003] Firearms training requires the use of targets to shoot at. To date these have primarily been paper targets which do not accurately represent real life. The paper targets are planar and do not react when shot in a manner that a real world target would.

[0004] US 2018/0051965 A1 discloses, according to its abstract, a modular shooting target for shooting practice having a first target component and a second target component, a trigger release element positioned adjacent to one of the first target component and the second target component and a connector element having a first end and a second end, the first end engaged with the trigger release element and the second end fixed to a secondary attachment point.

[0005] The trigger release element is a frangible element such as a clay rabbit or pigeon. A number of these clays are provided across the target, connected with the connectors. When one of the clays is shot, the target collapses. While this provides an element of real-world dynamics it is not especially accurate. The clays are relatively small and need to be directly hit to take down the target, while in reality the target area for a person is much larger. It also takes a long time to reset the target as the clays need to be replaced.

[0006] CN 217 210 585 U discloses, according to its abstract, a shooting training device, and relates to the technical field of shooting training equipment. This comprises a base and a rising and falling target arranged on the base, and the rising and falling target comprises a target rod and a motor for driving the target rod to rotate; one end of the target rod is connected with the output end of the motor, the other end of the target rod is provided with a target frame, and the target frame is provided with an electromagnet; the device further comprises a target changing mechanism, the target changing mechanism comprises a target bin and a jacking mechanism arranged in the target bin, a target body is arranged in the target bin, and magnetic metal is arranged on the target body. The controller is used for controlling the motor, the electromagnet and the jacking mechanism to cooperatively work so as to complete target replacement. According to the automatic target changing device, the automatic target changing function is achieved, the step of manual target changing is replaced, and the shooting training efficiency can be improved.

[0007] It is the changing of the target which is automated—i.e. the physical step of replacing one target with another. The damage on the target is observed in a conventional manner, and the user activates a trigger to replace the target once the target is too damaged or has received too many shots. This is a system for a conventional shooting range and does not replicate a real-world scenario.

[0008] US 2023/235996 A1 discloses, according to its abstract, a target shooting game. The target shooting game includes at least one frame, a plurality of targets, at least one actuator means, a controller means, and at least one power source. Further, preferably, the target shooting game includes a software application usable on an external device, such as a smartphone. The plurality of targets preferably include a gopher animal shape and are independently movable between a raised position and a lowered position, providing a user with a shooting game and/or shooting training session.

[0009] Again, this does not reflect a real-world training situation as the reaction of the targets is simply akin to a shooting gallery. The target does not fall vertically under gravity, but pivots about the hinge. This is not a realistic response.

[0010] There is therefore a need for an improved reactive target, which more accurately reflects

real-world operations.

SUMMARY

[0011] A reactive target for firearms training is provided. The reactive target comprising: a body; an impact sensor configured to detect an impact on a target region of the body; and an electromagnetic coupling arranged to support the body of the reactive target; and a controller configured to deactivate the electromagnetic coupling in response to the impact sensor detecting the impact for releasing the body of the reactive target to fall. Specifically the fall may be vertically under gravity. The fall can be unconstrained. The impact could be an impact from a projectile, and/or a shock wave impact such as from a percussion device like a stun grenade. Falling vertically under gravity can mean that there is no significant other force acting on the body during the fall, so it primarily falls strictly vertically under gravity.

[0012] The electromagnetic coupling may be arranged to selectively attach the body to a support structure. Selective attachment meaning that the coupling may attach and detach the body to and from the support structure. The support structure may be, for example, a stand, a frame, a wall, or the like.

[0013] The electromagnetic coupling may be arranged to support the body of the reactive target as its sole support. A sole support meaning that this is the only support for the body. Without this support, the body would fall—specifically fall vertically under gravity.

[0014] The body of the reactive target may be a mannequin comprising a torso. A mannequin being a 3D model of the human body. The torso of a mannequin refers to the central part that typically includes the chest, abdomen, and waist. The torso can also be identified as the upper body without limbs and/or a head.

[0015] The target region may be the torso of the mannequin. This generally corresponds to the “centre mass” which is a region that police and/or military forces are often taught to target since it can effectively incapacitate a threat.

[0016] The electromagnetic coupling may comprise an electromagnet and a magnetic section, the electromagnet attached to a head or the torso of the mannequin, or the magnetic section attached to the head or the torso of the mannequin. These are effective connection points to support the mannequin.

[0017] The mannequin may further comprise one or more limbs flexibly attached to the torso. This further increases the realism of the mannequin, and allows for the limbs to be posed. Further, forces applied to the limbs such as a projectile impact will be partially damped when transferred through to the torso.

[0018] The impact sensor may be for selectively detecting an impact on the one or more limbs in addition to the target region. By selectively it is meant that it is not certain that the impact on the limbs would be enough (due to the damping of the flexible connection) for the impact sensor and controller to deactivate the electromagnetic coupling. This more accurately reflects real life where an impact to the limbs could be enough to drop a target.

[0019] The impact sensor may be a vibration sensor or a tilt sensor. Such sensors can be particularly effective for identifying an impact.

[0020] The controller may be further configured to reactivate the electromagnetic coupling. This allows the reactive target to be quickly reset.

[0021] The controller may be configured to reactivate the electromagnetic coupling after a predetermined delay period. This means that the magnetic target is ready to reset automatically. For example, the predetermined delay period may be 0.1 seconds, 0.5 seconds, or 1 second.

[0022] The controller may be configured to deactivate the electromagnetic coupling in response to an impact on a target region with an impact force greater than a predetermined threshold. This helps ensure that only actual hits drop the reactive target.

[0023] The predetermined threshold may be adjustable. This means that the reactive target can be adjusted for different ammunition types and/or different difficulties.

[0024] The predetermined threshold may be selected for a type of projectile or percussion device, preferably for simunition or live ammunition. These are common projectile types that a user may wish to train with.

[0025] Live ammunition refers to rounds of firearms cartridges that contain functional and/or lethal components designed for use in live-fire situations. Unlike simunition or other training ammunition, live ammunition is loaded with traditional bullet projectiles, gunpowder, and/or primers.

[0026] Simunition refers to simulated ammunition. Simunition rounds are designed to mimic the look, feel, and/or ballistic properties of real ammunition. Simunition rounds often have with reduced velocity and/or energy compared to live ammunition.

[0027] The predetermined threshold may be selected such that an impact on the one or more limbs can be greater than the predetermined threshold. As noted above, having this as an option can further increase the realism of the reactive target.

[0028] The impact may be an impact of a projectile or percussion device. These are types of impact that would typically be desired to cause the target to react.

[0029] A kit is provided comprising the reactive target discussed herein, and a support structure, wherein: the electromagnetic coupling comprises an electromagnet and a magnetic section, and: the electromagnet is on the support structure and the magnetic section is on the reactive target; or the magnetic section is on the support structure and the electromagnet is on the body of the reactive target. This kit allows the reactive target to be set up anywhere.

[0030] The support structure may be a stand, frame, wall, or the like, These can be suitable support structures to support the reactive target.

[0031] The electromagnetic coupling may be a sole support attaching the body to the support structure. A sole support meaning that this is the only support for the body. Without this support, the body would fall—specifically fall vertically under gravity

[0032] A method of firearms training is provided. The method comprising: providing a reactive target as discussed herein; supporting the body of the reactive target using the electromagnetic coupling; impacting the target region of the body of the reactive target and sensing the impact with the impact sensor; and deactivating the electromagnetic coupling with the controller in response to the sensed impact to release the body of the reactive target to fall. Specifically the fall may be vertically under gravity. This method allows for effective firearms training as the reactive targets respond more like real world targets.

[0033] Supporting the body may comprise selectively attaching the body to a support structure. Selective attachment meaning that the coupling may attach and detach the body to and from the support structure. The support structure may be, for example, a stand, a frame, a wall, or the like.

[0034] The body of the reactive target may be solely supported using the electromagnetic coupling. That is, this step can be rewritten as solely supporting the body of the reactive target using the electromagnetic coupling.

[0035] The method may further comprise: reactivating the electromagnetic coupling with the controller; and re-supporting the body of the reactive target using the electromagnetic coupling. This allows the reactive target to be quickly reset.

[0036] The electromagnetic coupling may be reactivated after a predetermined delay period. This means that the reactive target is automatically ready to be reset.

[0037] The method may comprise comparing with the controller an impact force of the impact and deactivating the electromagnetic coupling if the impact force is greater than a predetermined threshold. This helps ensure that only actual hits drop the reactive target.

[0038] The method may further comprise: selecting the predetermined threshold based on a type of projectile or percussion device used for the method of firearms training, preferably for simunition or live ammunition. These are common projectile types that a user may wish to train with.

[0039] The body of the reactive target may be a mannequin comprising a torso. This is effective for

training with a target that reflects what the real world target may look like.

[0040] The target region of the body of the reactive target may be impacted with a projectile or percussion device. These are types of impact that would typically be desired to cause the target to react.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0041] The present specification makes reference, by way of example only, to the accompanying drawings in which:

[0042] FIG. 1 shows a reactive target.

DETAILED DESCRIPTION

[0043] A reactive target **100** for firearms training is shown in FIG. 1. This is specifically an example where the body of the reactive target is in the shape of a mannequin. That is, a 3D model of the human body.

[0044] In further examples the reactive target **100** could take and suitable shape. This could include a model of an animal, for example. Likewise, the mannequin could be a model of any humanoid-type of body-such as a monster/creature like a zombie or other mythological creature.

[0045] The reactive target **100** is called “reactive” because it reacts when it is hit by dropping, similar to a human target. This could also be called “interactive” or “realistic”.

[0046] The reactive target **100** has a body **1**. In the example of FIG. 1, the body is the mannequin shape. This body **1** has a target region. For example, this target region could be the torso **2** of the mannequin. The target region being the area of the body **1** which users of the reactive target **100** are expected to aim for and hit.

[0047] The body **1** can also comprise an ancillary region (such as the limbs discussed below in relation to a mannequin). The ancillary region may comprise a number of separate sub-regions. The ancillary region can correspond to other areas of the body **1** which users of the reactive target **100** are not expected to aim for but may still hit.

[0048] The ancillary region (or each individual sub-region) may be attached to the target region. This could be a flexible attachment. This means that there is an impact on the ancillary region, at least some of the impact force may be transferred to the target region. However, because the attachment is flexible there is a damping of the impact transferred.

[0049] The body **1** of the reactive target **100** is supported in a position for being shot at. For example with a mannequin this may correspond to a standing, crouching, or leaning position. The reactive target **100** further comprises an electromagnetic coupling **12** which supports the body **1** in this position. The electromagnetic coupling **12** comprises at least one electromagnet **12a**, and a magnetic section **12b**. In certain examples, the magnetic section **12b** may itself be a magnet and/or an electromagnet. The electromagnet **12a** contacts the magnetic section **12b** to form the coupling **12** which supports the body **1** of the reactive target.

[0050] One of the electromagnet **12a** or the magnetic section **12b** is provided on the body **1** of the reactive target **100**, and the other is provided on a support structure. For example the support structure could be one or more of a frame, stand, wall and/or ceiling.

[0051] The electromagnetic coupling **12** can be deactivated to release the body **1** of the reactive target **100**. By “deactivated” it means that the electromagnet **12a** is no longer provided with electricity and hence is no longer a magnet. This means that there is no magnetic force connecting the electromagnet **12a** and magnetic section **12b**. The electromagnet **12a** and magnetic section **12b** thus are detachable from one another under gravity.

[0052] FIG. 1 shows an example where the body **1** is a mannequin with a torso **2** and a head **4**. The head **4** generally corresponds to a human head. The electromagnet **12a** or magnetic section **12b**

may be provided on the head **4** and/or the torso **2** of the mannequin. It may be preferable for the electromagnet **12a** to be on the reactive target **100** since this means that power does not need to be provided to the magnetic section **12b** which may be for example on a frame, stand, wall and/or ceiling. For example, this could be on a back of the torso **2** so that the mannequin can be leant against a wall.

[0053] In certain cases, the electromagnetic coupling **12** may include a plurality of electromagnets **12a** and/or magnetic sections **12b**. This can allow the reactive target **100** to be attached in a number of positions and/or with multiple points of connection.

[0054] For example, the electromagnetic coupling **12** may suspend the target, or hold the target in an off-balance position. In certain cases, the electromagnetic coupling **12** may hold together a support within the reactive target **100**. Deactivating the electromagnetic coupling **12** can then release this support to allow the reactive target **100** to fall.

[0055] Specifically the fall may be vertically under gravity. That is, straight down and not pivoting about a hinge, for example. The fall can be unconstrained and/or unrestricted—or free.

Unconstrained in this context meaning that the movement of the fall is not restricted by another connection, such as a hinge. Other falling mechanisms are also possible in certain examples.

[0056] The reactive target **100** further comprises an impact sensor **16**. FIG. **1** shows this impact sensor **16** generally centrally, but it is appreciated that the impact sensor **16** can be arranged at any appropriate position on the body **1** of the reactive target **100**, particularly on the target region of the body **1**. The impact sensor **16** can be attached or otherwise affixed to the body **1** of the reactive target **100**.

[0057] The impact sensor **16** is configured so as to detect an impact on the target region of the body **1**. For example, impact onto the torso **2** of the mannequin of FIG. **1**. The impact could be an impact from a projectile, such as simunition or live ammunition. Additionally or alternatively, the impact could be a shock wave impact such as from a percussion device like a stun grenade, 9 bang, 3 bang, and/or flash bang. The shock wave passing through the air can be the impact on the target region of the body. Any references to an impact on the reactive target **100** in this specification must be interpreted as such. Particularly, these may be an impact with a projectile.

[0058] The impact sensor **16** may selectively detect an impact on the ancillary region—depending on how this impact is transferred to the target region.

[0059] The impact sensor **16** can in certain cases be by a vibration sensor, or a tilt sensor. However any suitable impact sensor **16** which can detect the impacts discussed herein may be used.

[0060] The reactive target **100** further comprises a controller **18**. The controller **18** can also be referred to as a processor. However, any form of controller **18** is possible. The controller **18** is in communication with the impact sensor to receive a signal indicative of an impact on the target region.

[0061] The controller **18** is further in electrical communication with the electromagnetic coupling **12**. The controller **18** is configured to deactivate the electromagnetic coupling **12** in response to the impact sensor **16** detecting an impact on the target region of the body **1**. Once deactivated, the body **1** is released and the body **1** is therefore able to fall. Specifically the fall may be vertically under gravity.

[0062] In this sense, the reactive target **100** is able to identify when a projectile (or other impact) has impacted the target region of the body **1**. In response to this, the reactive target **100** is released and falls, thereby simulating a real target being hit.

[0063] The controller **18** may deactivate the electromagnetic coupling **12** in response to an impact which has an impact force greater than a predetermined threshold. This could be as a result of the sensitivity of the impact sensor **16** being tuned such that a signal is only generated for an impact force beyond this predetermined threshold. Alternatively, or additionally, a signal may be generated for any impact and the controller **18** can then compare a magnitude of this signal against the predetermined threshold. If the impact is with less force than the predetermined threshold then the

electromagnetic coupling **12** may not be deactivated.

[0064] The predetermined threshold may be adjustable. For example, this could be with a user input. This can allow the reactive target **100** to be adjusted, for example for different types of projectile or percussion device which will impact with different forces. It also allows for sensitivity of the reactive target **100** to be adjusted. For example, if the predetermined threshold is increased then the impact may need to be nearer to the impact sensor **16** in order to reach this threshold, since the impact force would be dissipated about the rest of the reactive target **100**. This will be discussed in more detail below in relation to the arms **6** and legs **8**.

[0065] In certain cases, the controller **18** can also determine other characteristics of the impact in order to determine whether to deactivate the electromagnetic coupling **12**. For example, a duration of the impact may be compared a time threshold. An impact from a firearms projectile will be a short, sharp impulse. Therefore, if a length of the impact is greater than a time threshold the controller **18** may determine that the impact is not from a firearms projectile and hence not deactivate the electromagnetic coupling **12**.

[0066] It may be the case that the controller **18** can distinguish between percussion device impacts and projectile impacts. For example, this could be based on the predetermined threshold of impact force, and/or based on a profile of the impact (such as duration). In certain cases, the controller **18** may be configured to only deactivate the electromagnetic coupling **12** for a particular type of impact (i.e. only for projectile impacts).

[0067] This projectile can be any type of projectile which might be relevant for firearms training. This could include, for example, live ammunition and/or simunition.

[0068] Live ammunition refers to rounds of firearms cartridges that contain functional and/or lethal components designed for use in live-fire situations. Unlike simunition or other training ammunition, live ammunition is loaded with traditional bullet projectiles, gunpowder, and/or primers.

[0069] Simunition refers to simulated ammunition. Simunition rounds are designed to mimic the look, feel, and/or ballistic properties of real ammunition. Simunition rounds often have with reduced velocity and/or energy compared to live ammunition. Examples of simunition include wax bullets, rubber bullets, paintballs, etc.

[0070] It will be appreciated that different types of projectile may have different impact forces, as may different types of percussion devices. To this end, a predetermined threshold for the controller **18** can be adjusted based on the type of projectile or percussion device being used. The impact force can vary even within the types of ammunition. For example, simunition could include one or more of: airsoft pellets, paintballs, rubber bullets, and plastic bullets. On the other hand, live ammunition can be as varied as .22 caliber to .308 caliber, as well as shot gun slug and baton rounds.

[0071] The materials of the reactive target **100** can also be adjusted based on the type of projectile or percussion device being used. For example, a reactive target **100** for live ammunition may be constructed of ricochet-free material. For example, this could be ballistic rubber. Specifically this may be the body **1** of the reactive target **100**. This means that the live rounds pass through the reactive target **100**. In certain examples, no metal is used in the body **1** of the reactive target **100**.

[0072] In examples where the body **1** is a mannequin, this may further comprise one or more limbs—such as legs **6** and/or arms **8** shown in FIG. **1**. The limbs may be identified as the ancillary region discussed herein, and any reference to the limbs can equally be considered to refer to the ancillary region, or vice-versa.

[0073] These limbs **6**, **8** can be flexibly attached to the torso **2** of the mannequin. That allows the limbs **6**, **8** to be posed. For example, FIG. **1** shows how the arms **6** can be used to hold a firearm. Furthermore, if there is an impact on the limbs **6**, **8** at least some of the impact force may be transferred to the target region (i.e. to the torso **2**).

[0074] This means that it is possible for an impact on the limbs **6**, **8** to still be detected by the

impact sensor **16** and trigger deactivation of the electromagnetic coupling **12**. However, because the attachment is flexible there is a damping of the impact transferred. As a result, the force of the impact sensed by the impact sensor **16** may be below the predetermined threshold and hence the controller **18** does not deactivate the electromagnetic coupling **12**.

[0075] This can mean that impacts to the limbs **6, 8** are not reliable ways to ensure that the reactive target **100** falls. This is useful because it reflects real life where a shot to the torso **2** or head **4** may be more reliable for neutralising a threat than a shot to the limbs **6, 8**. Thus, having a shot to the limbs **6, 8** potentially able to deactivate the electromagnetic coupling **12** is more realistic.

[0076] In other words, the impact sensor **16** may selectively detect an impact on the limbs **6, 8**—depending on how the impact is transferred from the limbs **6, 8** to the torso **2**. Additionally, or alternatively, the predetermined threshold may be selected such that an impact on the limbs **6, 8** can be greater than this threshold.

[0077] The controller **18** may count a number of impacts. This could be a number of impacts greater than the predetermined threshold and/or a number of impacts less than the predetermined threshold. The controller **18** may deactivate the electromagnetic coupling **12** once a predetermined number of impacts have been counted. The predetermined number may be 2 or greater in certain cases. This predetermined number could be randomly determined by the controller **18**, for example at the start of an exercise. This could mean that different reactive targets **100** would need different numbers of such secondary impacts to fall.

[0078] For example, it may be that a number of impacts below the threshold (such as those on the limbs **6, 8**) could still trigger deactivation of the electromagnetic coupling **12**. This could reflect how in real life more hits may be needed in these regions to take down a threat. There could therefore be a first predetermined threshold which is lower than a second predetermined threshold. The second predetermined threshold corresponds to the predetermined threshold discussed herein where a single impact above this second predetermined threshold leads to deactivation of the electromagnetic coupling **12**.

[0079] An impact with an impact force greater than the first predetermined threshold may be counted by the controller **18**. Once the count of such impacts was greater than the predetermined number of impacts, the electromagnetic coupling **12** could be deactivated.

[0080] Of course, it is possible to have this counter used for all impacts to the reactive target **100** in examples where it is desired to have the reactive target **100** only fall after a number of hits to the target region.

[0081] In use, the reactive target **100** is set up by using the electromagnetic coupling **12** to attach the body **1** to a fixed point. For example and as noted herein that could be one or more of a frame, stand, wall and/or ceiling. Effectively, it could be any surface with a magnetic section **12b** attached thereto, or attachable thereto (or electromagnet **12a**).

[0082] Once the target region of the body **1** is impacted (i.e. shot), the controller **18** deactivates the electromagnetic coupling **12** and the body **1** falls under gravity. As noted above, deactivating the electromagnetic coupling **12** means that the electromagnet **12a** is no longer provided with electricity and hence is no longer magnetic. The electromagnetic coupling **12** can then be re-activated by providing the electromagnet **12a** with electricity such that it is a magnet again. The reactive target **100** can then be reset by re-contacting the electromagnet **12a** and the magnetic section **12b** to again support the body **1** of the reactive target **100**.

[0083] There may be a user input which triggers the reactivation of the electromagnetic coupling **12**. This may be via the controller **18**. Alternatively, or additionally, the controller **18** may reactivate the electromagnetic coupling **12** after a predetermined delay period, so that the reactive target **100** is ready to be reset quickly.

[0084] In certain cases, a kit of parts is provided for when the reactive target **100** is being used somewhere where there is not a fixed/permanent magnetic section **12b**. This kit includes the reactive target **100** as described herein, as well as a stand or frame. One of the magnetic section **12b**

or the electromagnet **12a** is provided on the stand, and the other is provided on the body **1** of the reactive target **100**. The stand can therefore support the body **1** of the reactive target **100**. For example, the stand may suspend the reactive target **100**. That is, the reactive target **100** may be hanging above the ground. Other ways of supporting the reactive target **100** could also be used.

[0085] This means that the reactive target **100** can be deployed wherever desired. A method of firearms training is also provided, which uses this reactive target **100**. The reactive target **100** can have any of the modifications discussed herein. This reactive target **100** is provided for the method.

[0086] The body **1** of the reactive target **100** is supported, using the electromagnetic coupling **12**. That is, the electromagnet **12a** and magnetic section **12b** may be brought into contact with one another such that the body **1** is supported. This could be a standing, crouching, or leaning position.

[0087] A trainee receiving the firearms training then impacts the target region of the body **1**. This may be an impact with a projectile, for example by shooting it. The projectile may be any suitable projectile, including live ammunition or simunition. Alternatively, or additionally, the impact could be with a percussive device. This impact is sensed by the impact sensor **16**.

[0088] Then, the controller **18** deactivates the electromagnetic coupling **12** in response to this impact. For example, this could be if the impact has an impact force greater than a predetermined threshold. The controller **18** may compare the impact force to this threshold (via a signal received from the impact sensor **16**), and/or the impact sensor **16** may be configured such that it only generates a signal if the impact force is greater than this threshold.

[0089] With the electromagnetic coupling **12** deactivated, the body **1** is released and the body **1** then falls under gravity. Specifically the fall may be vertically under gravity. This simulates a real target being hit.

[0090] In certain cases, the electromagnetic coupling **12** may be reactivated (i.e. provided with electricity to magnetize again). This may be via the controller **18**. The body **1** is then re-supported using the electromagnetic coupling **12**. That is, the electromagnet **12a** and magnetic section **12** may be brought back into contact with one another. This allows the reactive target **100** to be reset.

[0091] This reactivation of the electromagnetic coupling **12** could be after a predetermined delay period. For example, 0.1 seconds, 0.5 seconds, or 1 second, or any other time. This means that following a training exercise the reactive target **100** is already ready to be reset, thereby improving efficiency.

[0092] The method may include comparing with the controller **18** an impact force to a predetermined threshold, such as discussed herein. This threshold can be selected based on the type of projectile or percussion device being used, and could for example be selected to deactivate the electromagnetic coupling **12** when the target region of the body **1** is hit with simunition and/or live ammunition.

[0093] In certain examples, the method may include selecting, setting, or adjusting the predetermined threshold. This may be for example based on the type of projectile or percussion device being used.

[0094] While this method can be used for any sort of firearms training it may be particularly relevant when the body **1** of the reactive target **100** is a mannequin comprising a torso **2**, such as discussed herein. This mannequin may have any of the features discussed above.

[0095] In this sense, an improved reactive target **100**, kit, and method of firearms training is provided.

[0096] It will be appreciated that embodiments of the disclosure may be implemented using a variety of different information processing systems. In particular, although the Figures and the discussion thereof provide exemplary computing systems and methods, these are presented merely to provide a useful reference in discussing various aspects of the disclosure. Embodiments may be carried out on any suitable data processing device, such as a personal computer, laptop, tablet, personal digital assistant, mobile telephone, smart phone, set top box, television, server computer, etc. Of course, the description of the systems and methods has been simplified for purposes of

discussion, and they are just one of many different types of systems and methods that may be used. It will be appreciated that the boundaries between logic blocks are merely illustrative and that alternative embodiments may merge logic blocks or elements, or may impose an alternate decomposition of functionality upon various logic blocks or elements.

[0097] It will be appreciated that the above-mentioned functionality may be implemented as one or more corresponding modules as hardware and/or software. For example, the above-mentioned functionality may be implemented as one or more software components for execution by a processor/controller of the system. Alternatively, the above-mentioned functionality may be implemented as hardware, such as on one or more field-programmable-gate-arrays (FPGAs), and/or one or more application-specific-integrated-circuits (ASICs), and/or one or more digital-signal-processors (DSPs), and/or other hardware arrangements. Method steps implemented in flowcharts contained herein, or as described above, may each be implemented by corresponding respective modules. Moreover, multiple method steps implemented in flowcharts contained herein, or as described above, may be implemented together by a single module.

[0098] It will be appreciated that, insofar as embodiments of the disclosure are Implemented by a computer program, then a storage medium and a transmission medium carrying the computer program form aspects of the disclosure. The computer program may have one or more program instructions, or program code, that, when executed by a computer, causes an embodiment of the disclosure to be carried out. The term “program” as used herein, may be a sequence of instructions designed for execution on a computer system, and may include a subroutine, a function, a procedure, a module, an object method, an object implementation, an executable application, an applet, a servlet, source code, object code, a shared library, a dynamic linked library, and/or other sequences of instructions designed for execution on a computer system. The storage medium may be a magnetic disc (such as a hard drive or a floppy disc), an optical disc (such as a CD-ROM, a DVD-ROM or a Blu-ray disc), or a memory (such as a ROM, a RAM, EEPROM, EPROM, Flash memory or a portable/removable memory device), etc. The transmission medium may be a communications signal, a data broadcast, a communications link between two or more computers, etc.

[0099] Each feature disclosed in this specification, unless stated otherwise, may be replaced by alternative features serving the same, equivalent, or similar purpose. Thus, unless stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

[0100] As used herein, including in the claims, unless the context indicates otherwise, singular forms of the terms herein are to be construed as including the plural form and, where the context allows, vice versa. For instance, unless the context indicates otherwise, a singular reference herein including in the claims, such as “a” or “an” (such as a component or an element) means “one or more” (for instance, one or more components, or one or more elements). Throughout the description and claims of this disclosure, the words “comprise”, “including”, “having” and “contain” and variations of the words, for example “comprising” and “comprises” or similar, mean that the described feature includes the additional features that follow, and are not intended to (and do not) exclude the presence of other components.

[0101] The use of any and all examples, or exemplary language (“for instance”, “such as”, “for example” and like language) provided herein, is intended merely to better illustrate the disclosure and does not indicate a limitation on the scope of the disclosure unless otherwise claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the disclosure.

[0102] Any steps described in this specification may be performed in any order or simultaneously unless stated or the context requires otherwise. Moreover, where a step is described as being performed after a step, this does not preclude intervening steps being performed.

[0103] All of the aspects and/or features disclosed in this specification may be combined in any combination, except combinations where at least some of such features and/or steps are mutually

exclusive. In particular, the preferred features of the disclosure are applicable to all aspects and embodiments of the disclosure and may be used in any combination. Likewise, features described in non-essential combinations may be used separately (not in combination).

[0104] A method of manufacturing and/or operating any of the devices disclosed herein is also provided. The method may comprise steps of providing each of the features disclosed and/or configuring or using the respective feature for its stated function.

Claims

1. A reactive target for firearms training, the reactive target comprising: a body; an impact sensor configured to detect an impact on a target region of the body; and an electromagnetic coupling arranged to support the body of the reactive target; and a controller configured to deactivate the electromagnetic coupling in response to the impact sensor detecting the impact for releasing the body of the reactive target to fall vertically under gravity.
2. The reactive target of claim 1 wherein the electromagnetic coupling is arranged to selectively attach the body to a support structure.
3. The reactive target of claim 1, wherein the electromagnetic coupling is arranged to support the body of the reactive target as its sole support.
4. The reactive target of claim 1, wherein the body of the reactive target is a mannequin comprising a torso.
5. The reactive target of claim 4, wherein the target region is the torso of the mannequin.
6. The reactive target of claim 4, wherein the electromagnetic coupling comprises an electromagnet and a magnetic section, the electromagnet attached to a head or the torso of the mannequin, or the magnetic section attached to the head or the torso of the mannequin.
7. The reactive target of claim 4, wherein the mannequin further comprises one or more limbs flexibly attached to the torso.
8. The reactive target of claim 7, wherein the impact sensor is for selectively detecting an impact on the one or more limbs in addition to the target region.
9. The reactive target of claim 1, wherein the impact sensor is a vibration sensor or a tilt sensor.
10. The reactive target of claim 1, wherein the controller is further configured to reactivate the electromagnetic coupling.
11. The reactive target of claim 10, wherein the controller is configured to reactivate the electromagnetic coupling after a predetermined delay period.
12. The reactive target of claim 1, wherein the controller is configured to deactivate the electromagnetic coupling in response to an impact on a target region with an impact force greater than a predetermined threshold.
13. The reactive target of claim 12, wherein the predetermined threshold is adjustable.
14. The reactive target of claim 12, wherein the predetermined threshold is selected for a type of projectile or percussion device, preferably for simunition or live ammunition.
15. The reactive target of claim 12, when dependent on claim 5 or 6, wherein the predetermined threshold is selected such that an impact on the one or more limbs can be greater than the predetermined threshold.
16. The reactive target of claim 1, wherein the impact is an impact of a projectile or percussion device.
17. A kit comprising the reactive target of claim 1, and a support structure, wherein: the electromagnetic coupling comprises an electromagnet and a magnetic section, and: the electromagnet is on the support structure and the magnetic section is on the reactive target; or the magnetic section is on the support structure and the electromagnet is on the body of the reactive target.
18. The kit of claim 17, wherein the support structure is a stand, frame or wall.

- 19.** The kit of claim 17, wherein the electromagnetic coupling is a sole support attaching the body to the support structure.
- 20.** A method of firearms training, comprising: providing a reactive target according to claim 1; supporting the body of the reactive target using the electromagnetic coupling; impacting the target region of the body of the reactive target and sensing the impact with the impact sensor; and deactivating the electromagnetic coupling with the controller in response to the sensed impact to release the body of the reactive target to fall vertically under gravity.
- 21.** The method of claim 20, wherein supporting the body comprises selectively attaching the body to a support structure.
- 22.** The method of claim 20, wherein the body of the reactive target is solely supported using the electromagnetic coupling.
- 23.** The method of claim 20, further comprising: reactivating the electromagnetic coupling with the controller; and re-supporting the body of the reactive target using the electromagnetic coupling.
- 24.** The method of claim 20, wherein the electromagnetic coupling is reactivated after a predetermined delay period.
- 25.** The method of claim 20, wherein the method comprises comparing with the controller an impact force of the impact and deactivating the electromagnetic coupling if the impact force is greater than a predetermined threshold.
- 26.** The method of claim 25, further comprising: selecting the predetermined threshold based on a type of projectile or percussion device used for the method of firearms training, preferably for simunition or live ammunition.
- 27.** The method of claim 20, wherein the body of the reactive target is a mannequin comprising a torso.
- 28.** The method of claim 20, wherein the target region of the body of the reactive target is impacted with a projectile or percussion device.
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